

To Facilitate or not to Facilitate: Human and LLM Facilitator Tendencies in Online Discussions

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Abstract

Automating facilitation in online discussions is a long-standing social concern given the increasing time we spend on online spaces and the failure of content moderation approaches. While studies have been conducted on how to facilitate, none have answered the essential question of when to do so. A potential answer is using LLMs, which ostensibly make automated, large-scale intervention increasingly feasible. In this study, we examine when LLMs decide to facilitate by defining what facilitation is, observing when humans decide to facilitate, and comparing their decisions with those made by LLMs. To this end, we create PEFK, a corpus standardizing and aggregating all relevant facilitation datasets. We are the first to run a survey on facilitation timing, which we execute using expert facilitative participants and LLM-as-a-judge models. We discover that while humans are more cautious, LLMs are excessively eager to facilitate, although both are more certain when judging that facilitation is not needed. We then investigate whether this behavior can be corrected using alternative setups for LLMs and training ModernBert classifiers on established datasets, finding that the latter perform more reliably than the former, although current datasets impose a relatively low performance ceiling.

1 Introduction

Given the scale of social media networks and online spaces, facilitation needs to be automated to keep up with the amount of posted content (Small et al., 2023; Burton et al., 2024). So far, platforms have largely ignored the issue, preferring automated flag-and-remove mechanisms (“content moderation”) (Korre et al., 2025), which are not sufficient (Cresci et al., 2022; Trujillo and Cresci, 2022). While recent work has undertaken the task of choosing *how* to facilitate (Chen et al., 2025; Schroeder et al., 2024; Park et al., 2012), there has been little interest in *when* to do so (Falk et al., 2021).

Deciding when to intervene in a discussion is a matter of great importance. Intervening too early or too frequently may lead to users expressing frustration at being constantly interrupted, while intervening too late

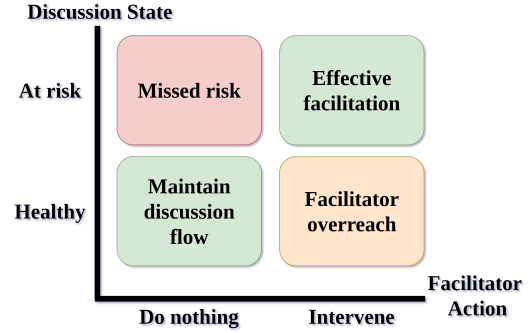


Figure 1: At any point in a discussion, the facilitator must decide whether to intervene or not. Not intervening when the situation calls for it may lead to the discussion worsening (e.g., topic derailment, escalation, missed participation). On the other hand, intervening when the discussion is already going well may lead to irritation among participants.

may lead to a significant escalation and derailment of the discussion (Fig. 1, Korre et al. (2025); Gao et al. (2025)). Due to the sheer scale of modern online spaces automation is needed, which the field of Natural Language Processing (NLP) has attempted to implement using models to determine when an intervention is necessary. Encoder-Only (EO) classifiers were used to this end (Falk et al., 2021), although recently LLMs have been championed as the eventual solution to online facilitation (Korre et al., 2025; Small et al., 2023). However, previous work in LLM facilitation indicates that LLM facilitators are way too eager to intervene, rendering them unusable as autonomous facilitative agents (Tsirmpas et al., 2026). Yet, this finding has not been contrasted with human tendencies. In this study, we focus on the following Research Question (RQ):

RQ. *How and why do LLMs differ with humans when deciding when to facilitate?*

In order to answer the central research question we need to answer the following sub-questions:

What constitutes a facilitative intervention? There is currently no consensus on how to define a facilitative intervention. By examining relevant literature, we find three common definitions implicitly used in prior work; *escalation*, *professional facilitation*, and *facilitative comments* (§2.1).

Where do humans facilitate? While the majority of content where facilitation is needed is posted on online, written discussions, recent work predominantly focuses on *domain-specific oral discussion* scenarios (e.g., radio-hosted debates—see App. B.3). Since datasets for discussion facilitation are generally few, small, and straddle radically different domains (e.g., debates, forums, deliberations), we gather all datasets relating to any of the three possible definitions of the term and compile the “Prosocial and Effective Facilitation in Konversations (PEFK)”, a standardized dataset for discussion facilitation with a unified data and label schema (§3). Our work thus attempts to transfer knowledge from specialized, oral datasets originating from different domains, to the important field of (written) online facilitation.

How do humans facilitate? We create and execute a survey task designed to gauge human facilitative patterns by sampling 1226 discussion chunks from our dataset, and asking 10 expert facilitative participants whether they would intervene as facilitators. We find that human participants tend to take a cautious approach to facilitation, generally preferring not to intervene and expressing greater confidence in not doing so (Finding 1). Additionally, human participants may pay attention to different spans of a discussion, which can lead them to choose different courses of action regarding facilitation.

How does LLM facilitation differ? We repeat the survey with six open-source LLMs. Unlike humans, LLMs tend to over-intervene, confirming previous studies (Tsirmpas et al., 2026) and rely more heavily on positive reinforcement. However, compared to humans, they may be more consistent between themselves, hinting that they are likely focusing on similar spans of the discussion. We also find that, much like humans, determining when not to facilitate is easy—the opposite is not (Findings 2,3).

How can we fix LLM facilitation? We investigate whether aggregating all currently available datasets can support either the training of classifiers or the use of out-of-the-box LLMs for a simpler and more commonly studied task in the literature: predicting whether a facilitator would intervene, rather than whether a facilitative comment is needed (§5). Our results show that LLMs are less reliable than ModernBert (Warner et al., 2025) classifiers, although their performance depends heavily on the specific model (Finding 4). We also observe a relatively low upper bound on performance, caused from the inherent noise introduced by defining facilitative comments as those written by professional facilitators (Finding 5).

Overall, our study offers the following contributions: We bridge differences between prior work on (i) what a facilitation intervention is, (ii) oral and written discussions (1), by compiling the largest standardized facilitation dataset so far (2). Through the execution of an extensive survey performed by experts on facilitation

(3), we show that the key difference between humans and LLMs rests in how frequently they deem facilitation to be necessary (4). We are the first to evaluate LLMs for facilitation timing prediction, finding that they are wildly inconsistent while EO models offer a more reliable alternative (5), though both approaches remain constrained by the limited current understanding of facilitation within NLP (6). Ultimately, achieving reliable automated facilitation requires moving beyond surface signals and using comprehensive, human-informed data that captures the true complexity of human facilitation. We release the code for the creation of our dataset and the experiments presented in this paper under a GPLv3-or-later license.¹

2 Related Work and Background

2.1 What are facilitative interventions?

Facilitation (or conversational moderation) refers to the active participation of people in order to improve discussions (Korre et al., 2025). Relevant work assumes that facilitation is conducted by specially trained staff (Schroeder et al., 2024; Park et al., 2012; Chen et al., 2025), or users appointed by the community (Schaffner et al., 2024). Studying relevant literature on facilitation yields multiple definitions for what a facilitative intervention is, which can be categorized to three main definitions:

Definition 1: Escalation. *A facilitator needs to intervene at the precise point at which, without facilitation, the discussion would derail or escalate.*

Escalation is usually measured as the point where the participants request arbitration (e.g., escalate the matter to a site moderator in Wikipedia (De Kock and Vlachos, 2021; Chang and Danescu-Niculescu-Mizil, 2019)). This definition is quite appealing for NLP researchers as the labels are easy to acquire since a ground truth exists (i.e., “escalation” can be defined in terms of moderator and user actions specific to each platform), and the existence of a single, largely unambiguous ground-truth is ideal for the training and evaluation of any Machine Learning (ML)-based classifier. However, it is also inherently restrictive, as a facilitator has multiple duties that do not directly relate to preventing personal attacks or rhetorical escalation such as summarizing discussions (Small et al., 2023), managing participation (Chen et al., 2025; Schroeder et al., 2024), and providing important information for new members (Park et al., 2012). Furthermore, the datasets require that the platforms contain an unambiguous signal that a discussion has escalated, which is not often the case (Table 2). Thus, this definition is not considered in this study, but is included in our dataset (§3).

¹Dataset creation: <https://github.com/dimits-ts/facilitation-dataset>
Experiments code: <https://github.com/dimits-ts/interventions>
Survey data preprocessing: Available upon request.

Definition 2: Professional facilitation. *Facilitation should occur at the point where a professional facilitator would have intervened.*

In current literature, it is generally accepted that all comments made by a facilitator should be considered facilitative interventions; for instance, only these comments are assumed to follow some facilitative strategy (Falk et al., 2021; Chen et al., 2025; Schroeder et al., 2024). Defining facilitative comments this way is simple and intuitive, but can only be used in domains where (1) professional facilitators exist and (2) they can be identified. These conditions exclude the vast majority of online discussions, as very few web-crawled datasets include such information. This can be seen clearly in Table 2, where the largest datasets such as WikiConv (Hua et al., 2018) cannot be used for analyzing facilitation using this definition, since there is no explicit information on which user is a facilitator (in the case of WikiConv, a verified Wikipedia moderator).

Definition 3: Human judgment. *Facilitation should only occur when the majority agree that it is necessary.*

There is an argument to be made that professional facilitators can make more accurate judgments due to expertise. However, in reality most facilitation is handled by either volunteers elevated to the position by their community (Schaffner et al., 2024) or by the users themselves in an ad-hoc fashion (Falk et al., 2024). In this case, the task is no longer about detecting or predicting whether a facilitator would speak, but rather the much more difficult and subjective task of whether the comment is *facilitative in nature*.

What is the difference? Table 1 shows an example where the same discussion is annotated following the three definitions. Def. 1 is reliant on the outcome of the discussion, and would miss examples where successful interventions led to a smooth discussion. Def. 2 would give more fine-grained information, although we run the risk of classifying non-facilitative comments as facilitative (and sometimes the opposite (Falk et al., 2024)). Finally, Def. 3 can provide us with fine-grained information, although ambiguous cases remain.

2.2 How to automatically determine whether to facilitate

Using humans Recent research states that a fully-automated facilitation is still out of our reach as LLMs tend to be very intruding and repetitive, and fixate on the moderation side of facilitation (Tsirmpas et al., 2026; Korre et al., 2025). Expert human facilitators are frequently used in research (Chen et al., 2025; Park et al., 2012; Schroeder et al., 2024), but this is not a scalable process due to costs (Rossi et al., 2024).

Using models While automated content moderation is widespread on social media platforms, automated facilitation is a topic currently exclusive to academic research (Korre et al., 2025). The only work to our

Name	Text	D1	D2	D3
Joe	Thank you for sharing. So I am Joe from Chicago originally...	×	×	×
Marie	Thank you Joe.	×	✓	?
Peter	Yeah. So I'll kind of go through it again. I am Peter...	×	×	×
Marie	Thank you, Peter. So I think we have another participant on the line...	×	✓	✓

Table 1: Example text where facilitation interventions are identified according to each definition (D1-3). *Marie* is a facilitator. We assume the discussion does not derail. The second comment *may* be facilitative according to Def. 3, which depends on the consensus of participants.

knowledge that has explicitly attempted to predict facilitative interventions (Falk et al., 2021) uses EO classifiers on a single dataset—the Cornell eRulemaking Initiative (CeRI) dataset (Park et al., 2012)—with modest performance. Indeed, these classifiers have proven competent in the context of escalation prediction (Chang and Danescu-Niculescu-Mizil, 2019; Yuan and Singh, 2023), and facilitative tactics detection (Chen et al., 2025; Schroeder et al., 2024). Recent work (Schroeder et al., 2024; Chen et al., 2025) uses LLM-as-a-judge models to annotate data in the field of facilitation, although these models may not be able to be used directly in facilitation tasks (Tsirmpas et al., 2026).

Ignoring it Most platforms hold users responsible for maintaining the quality of discussion and the enforcement of community and platform rules, especially in jurisdictions where the platforms themselves are largely absolved of the responsibility of posted content, such as the U.S. (Seering, 2020; Schaffner et al., 2024). In some cases, the users are responsible for just flagging and reporting content (Seering, 2020), while platforms like Reddit use volunteer moderators, who also function as community leaders and facilitators (Seering, 2020; Falk et al., 2024), raising ethical questions about “Moderation as Free Labor” (Matias, 2019). The indifference of such platforms towards facilitation can be attributed as one of the main reasons that datasets on facilitation are few, small, and fragmented across different domains and facilitation sub-tasks.

3 PEFK: A large, multi-domain facilitation dataset

Since there is currently no dataset that explicitly concerns itself with facilitation interventions, we construct the “Prosocial and Effective Facilitation in Konversations (PEFK)” dataset, a standardized, multi-domain compilation of the facilitation-related datasets listed by Korre et al. (2025) (see Table 2). PEFK contains data from both online forums and oral (e.g., debates, radio) discussions, and standardizes metadata, metrics, and labeling schemes according to a simple, unified schema,

Name	#Comments	#Discussions	Domain	Format	Fac.
WikiDisputes (2021)	96,320	8,727	Forum	Text	×
WikiTactics (WT) (2022)	3,850	213	Forum	Text	✓
WikiConv (2018)	17,806,373	4,486,983	Forum	Text	×
Conversations Gone Awry / CMV II (2019)	40,607	6,691	Forum	Text	×
Cornell eRulemaking Initiative (CeRI) (2012)	3,700	132	Forum	Text	✓
User Moderation (UMOD) (2024)	2,000	1,000	Forum	Text	✓
Intelligence Squared (IQ2) (2016)	34,245	108	Formal debate	Oral	✓
Why-How-Who (WHoW) (2025)	25,542	178	Radio / TV	Oral	✓
Fora (2024)	39,438	262	Story sharing	Oral	✓

Table 2: The datasets used in PEFK. We include a large variety of domains, and two different sources; text-only datasets, and transcripts from oral discussions. All datasets include multi-participant discussions (besides UMOD which contains comment pairs). The **Fac.** column denotes whether information on which users are facilitators is available.

making it suitable as a benchmark on facilitator interventions. More information on preprocessing steps and dataset schema can be found in App. B.

PEFK is designed to facilitate exploration and analysis of facilitative interventions under all three definitions described in §2.1. Specifically, we use a subset of the data that includes professional facilitation annotations to train our models according to Def. 2 in §5 (see App. B.3; Fig. 5). We also use a subset of PEFK to survey human facilitative patterns according to Def. 3, although the responses are not provided as ground-truth in the dataset itself. Finally, we include escalation and toxicity, as used in Def. 1, although this information is not utilized in this study (see §2.1).

4 Do humans and LLMs intervene at similar circumstances?

To support a systematic investigation of facilitative behavior under Def. 3, we conduct a survey on PEFK to analyze facilitation timing patterns between humans and LLMs. Rather than establishing a ground truth through Inter-Annotator Agreement (IAA), we adopt a survey-based approach, since prior work has shown that obtaining reliable annotations for similar tasks is highly challenging, particularly in newly explored settings such as ours (see “Survey Procedure”).

4.1 Facilitative survey

Labels To avoid turning facilitation into a moderation task (i.e., using reactive responses after unwanted comments/utterances), we introduce two types of interventions: positive and negative reinforcement. *Positive reinforcement*, aims to encourage or reward constructive behavior (e.g., expressing appreciation or affirming contributions), while *negative reinforcement*, aims to discourage undesirable behavior or guide interlocutors back to productive discourse (e.g., addressing incivility or off-topic comments). This distinction represents a tool through which we ensured the survey participants did not default to content moderation, which we frequently observed during discussions with them, and during pilot survey rounds—it does not represent a theoretical contribution. Nevertheless, it provides an extra

signal which we utilize in our analysis.² We adopt an ordinal scale approach, where the survey participants rate each dialogue for either type of reinforcement (positive or negative), as well as the possibility of no reinforcement using a scale from 1 to 5.³

Data We randomly split multi-party discussions from the PEFK dataset into small chunks containing a set amount of comments (6 comments per discussion), and select a random sample for the survey (1224 discussions—721 remained after the participants flagged some excerpts as malformed). This design aims to replicate a known limitation in online asynchronous facilitation scenarios; the facilitator is expected to decide on a course of action considering only the last parts of a discussion, as shown by several studies (Hewitt, 2003; Park et al., 2016; Kumpel and Unkel, 2021) and a crowdsourced moderation platform (Lampe et al., 2014). While this justification does not apply directly to oral discussions, there is evidence to suggest that the most recent comments in a discussion are given relatively greater importance (Crano, 1977; Cockburn et al., 2017).

Survey Participants We recruited 11 experts on Computational Social Science⁴, and who had extensive experience with NLP tasks from sentiment analysis to persuasion strategy detection. Additionally, before the survey, they were given a training task where they categorized facilitative tactics according to established taxonomies on PEFK (see App. D.1) and given ample resources and time to familiarize themselves with online facilitation. The gender distribution was relatively balanced with 4 males and 7 females, while their ages had a median

²There is some theoretical grounding for using this approach is described in App. D.2. Similarly, Falk et al. (2021) use high and low Argument Quality for further analysis of facilitation prediction, although, unlike their approach, we obtain the explanatory signal directly from the survey participants, rather than established EO models.

³In order to transform the multiple ordinal labels into an easily manageable form for the purposes of analysis, we transform them into a single multiclass label schema as described in App. D.3.

⁴One participants (A1) did not complete the task and was therefore not considered for the analysis.

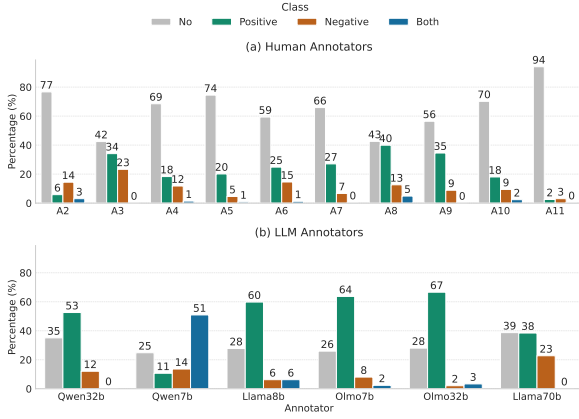


Figure 2: Distribution of all classes per participant. Human participants are either very cautious (A2, A4–A7, A10, A11) or somewhat cautious (A3, A8, A9). On the other hand, the LLMs (b) are significantly more eager to facilitate than humans (a).

of 30 years. Before the commencement of the survey, the participants completed an agreement form in compliance with GDPR (European Union, 2016) ethical requirements and were compensated in accordance with national legislation.

Survey Procedure Unlike prior work that sought to establish consensus among facilitators (Falk et al., 2024; Schroeder et al., 2024; Chen et al., 2025), our goal is instead to explore facilitation patterns without presuming that such a consensus exists. Indeed, facilitation is a highly subjective task, and substantial disagreements persist even among experienced professionals operating within the same communities and under the same rule-sets (Koshy et al., 2024); consensus remains elusive even in cases where the task is less ambiguous and the dataset more focused (Schroeder et al., 2024).⁵ The task required the participants to rate instances from PEFK by determining whether, at any point within a specific discussion chunk, reinforcement was needed, as well as what kind. The participants could optionally add rationales with which they could justify their rating. The full survey guidelines can be found in App. D.

LLM Survey We use a variety of open-source LLMs (see §C.1) to ensure reproducibility. The prompts used were adopted from the guidelines provided to the experts—see App. E.2.1.

4.2 Human facilitation

Humans make cautious facilitators According to Fig. 2, facilitation preferences vary across participants but generally remain conservative. Most participants choose not to intervene in most cases, although some (e.g., A3, A8, and A9) exhibit comparatively more

⁵During the survey, we inserted duplicated entries in secret to estimate consistency, finding that all-but-one participants remained 100% consistent in their judgments during the task—see App. D.5.

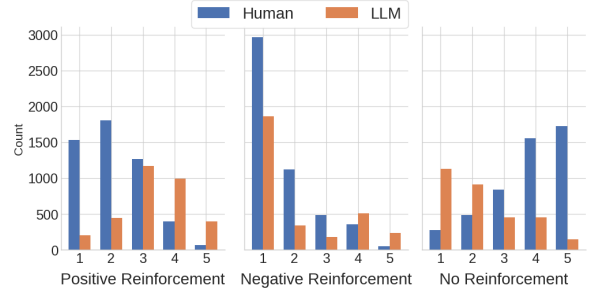


Figure 3: Certainty of judgments per facilitation label. LLMs (orange bars on the right) are much more confident in their judgments than humans (high frequencies on the extreme modes i.e., 1 and 5).

#	Turn
1	Well nothing is really *inherently* bad...
2	Well political correctness is about avoiding offense...
4	Are you really arguing that WWI was unjust towards cisgendered men?
5	Not WWI itself, but that period, and I don't see what is so odd about it...
6	There were transgendered people in WWI. You still haven't justified how it was genuinely an injustice...

Table 3: Sections from the same discussion chunk. Participants facilitate differently depending on which section they decide is more important.

proactive behavior with higher rates of positive interventions. Positive-intervention rates also appear to cluster around two broad ranges (approximately 20–30% and 35–40%), which is consistent with the tendencies of professional facilitators observed in the original dataset (see Fig. 5; App. B.2).⁶ After the conclusion of the survey we realized that participants should provide detailed justifications for their labels. While we treated such information as optional, we found that they significantly improve the interpretability of our results.

Human facilitators may attend over different discussion spans To qualitatively understand the sources of divergence in human facilitative patterns, we analyzed a “worst-case” scenario: situations where half of the participants favored positive reinforcement and the other half favored negative reinforcement. As shown in Table 3, a specific discussion example illustrates this conflict. Participants justifying negative reinforcement cited the conversation’s trajectory toward escalation, suggesting intervention was necessary to curb aggressive language. Conversely, the same participants noted that certain comments—particularly comments 1 and 2—were examples of politeness and hedging, making them candidates for positive reinforcement. This case demonstrates how humans weigh conflicting signals within a single discussion.

Finding 1. Most human participants tend to be either

⁶These observations are consistent across datasets—see App. D.4—reinforcing our motivation of transferring observations from oral datasets to online, written facilitation.

706	Pos	Both	Pos	Pos	Pos	Pos	No	No	No	No	No	No	No	No	No	No	No
582	Pos	Both	Pos	Pos	Pos	Pos	Neg	No	No	Neg	Neg	No	Both	Neg	Neg	Neg	Neg
520	Pos	Both	Pos	Pos	Pos	Pos	No	Neg	No	No	No	No	No	No	No	No	No
459	Pos	Both	Pos	Pos	Pos	Pos	No	Neg	No	No	No	No	Both	No	No	No	No
661	Pos	Both	Pos	Pos	Pos	Pos	Neg	No	No	No	No	No	No	No	Neg	No	No
Qwen32b	Qwen7b	Llama3b	Qwen7b	Qwen32b	Llama70b	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11		

Figure 4: Top 5 discussion instances where the LLM majority vote disagrees with all human participants. The y-axis indicates the discussion instance index in the dataset. Labels follow the survey schema described in App. D.2: POS = Positive Reinforcement, NEG = Negative Reinforcement, BOTH = Both Positive and Negative Reinforcement, and NO = No Reinforcement.

highly cautious or to balance proactivity with caution. Disagreements may arise because participants focus on different spans of the discussion, each of which may call for distinct forms of reinforcement, highlighting the importance of including rationales in the survey process.

4.3 Do LLMs facilitate differently?

The facilitative style of LLMs differs significantly

An examination of the distribution of each possible value per class (Fig. 3) shows that humans are less certain than LLMs when it comes to positive reinforcement but in both cases the values are concentrated in the middle part of the distribution. For negative reinforcement, both humans and LLMs show high uncertainty, while for no reinforcement there is a clear difference with humans being more certain than LLMs.

Why exactly do LLMs differ from humans? For each comment, we compute the LLM majority label and calculated the intra-LLM agreement fraction. Filtering for comments with zero human agreement, we identified the most extreme cases, where LLMs largely agreed among themselves but completely disagreed with all humans. This analysis highlights situations where model behavior diverges acutely from human judgment, showing again that LLMs are not in a condition to be used in real-life settings. In total, there are 36 cases where no human agrees with the majority voting of the LLMs. Fig. 4 shows the top 5 extreme cases. Despite strong LLM consensus (agreement ≥ 0.66), all human participants labeled these comments differently.

Unfortunately, we cannot rely on the LLMs for explanations post-hoc, nor use Chain of Thought (CoT) prompting to extract rationales, since both produce non-representative and deceptive explanations (Sanwal, 2025; Barez et al., 2025; Turpin et al., 2023). How the models arrived to these conclusions must therefore be evaluated from our own observations. Examining the five examples qualitatively, we see that all cases regard a narrative of the participants rather than a debate like discussion that could possibly lead to escalation. Specifically, discussion 706 is about gender, labor market change, and education, where multiple speak-

ers exchange viewpoints on shifting economic roles.⁷ The excerpt contains a lot of anecdotal examples (e.g., references to one’s workplace examples and “Aviation High School”), rather than decision-oriented, focusing on broad social interpretation. In general, the dialogue is very self-contained, with speakers building perspectives rather than seeking resolution or guidance. This narrative like structure lead to the participant to choose to not to intervene and reporting that discussions like these comprise “neutral comments”. LLMs, on the other hand, being more prone to intervention, could interpret this neutrality as an opportunity for positive reinforcement. This is a hypothesis that should be verified with rationales by the participants in future research.

Finding 2. *LLMs are more intrusive and more consistent compared to humans, being eager to view neutral discussions as opportunities for positive reinforcement.*

5 Are existing datasets useful?

As discussed in §2.1, facilitation interventions can be defined in two ways; whether a comment is facilitative in nature (Def. 3), or whether a dedicated facilitator would participate at a given point in time (Def. 2). In this section, we investigate whether we can use available datasets to train and evaluate models using the latter definition.

5.1 Experimental setup

Def. 2 and Def. 3 are radically different, and thus require completely separate methodologies (see §2). We frame the task as a classification problem, aiming to identify the last comment made by a non-facilitator before a facilitator intervenes. To this end, we evaluate six open-source LLMs (App. C.1) and train a Modern-Bert classifier (Warner et al., 2025).⁸ Participant names are redacted, and each instance includes the three preceding comments as context. We use a 60–20–20 split for training the classifier, and choose 2,000 randomly sampled instances from the same test set for the LLM evaluation, evenly distributed across the sub-datasets of PEFK. We use a single Quadro RTX 6000 GPU for the classifier and two GPUs for the LLMs, collectively requiring about 72 hours of computation; see further details in App. C.2.1.

5.2 The easy choice: deciding when NOT to facilitate

Table 4 shows the results for the facilitator intervention prediction task. We find that while most models are adept at identifying cases where facilitators did not intervene (large $F1_n$), they are generally much less

⁷Since each discussion span can involve as many as 30,000 characters, we refer the reader to the supplementary material for context.

⁸Due to substantial differences in text format, we train separate models for oral and written datasets (App. B.3); an ablation study using a single model across all datasets shows reduced performance (App. C.2.2).

Model	Pr.	Rec.	$F1_p$	$F1_n$	Sup.
CeRI					
LLaMa70B	0.152	0.082	0.107	0.783	335
LLaMa8B	0.246	0.518	0.333	0.567	335
ModBert	0.319	0.490	0.386	0.706	192
OLMo32B	0.154	0.071	0.097	0.795	335
OLMo7B	0.251	0.988	0.401	0.000	335
Qwen32B	0.100	0.024	0.038	0.821	335
Qwen7B	0.182	0.118	0.143	0.774	335
All	0.201	0.327	0.215	0.635	335
WT					
LLaMa70B	0.241	0.433	0.310	0.310	336
LLaMa8B	0.328	0.842	0.472	0.074	336
ModBert	0.407	0.717	0.520	0.613	258
OLMo32B	0.292	0.466	0.359	0.465	331
OLMo7B	0.348	0.958	0.511	0.009	336
Qwen32B	0.267	0.258	0.263	0.601	336
Qwen7B	0.264	0.392	0.315	0.455	336
All	0.307	0.581	0.393	0.361	336
Fora					
LLaMa70B	0.333	0.058	0.099	0.760	382
LLaMa8B	0.386	0.584	0.465	0.562	382
ModBert	0.409	0.631	0.497	0.599	2764
OLMo32B	0.391	0.066	0.113	0.765	382
OLMo7B	0.358	0.993	0.526	0.008	382
Qwen32B	0.438	0.051	0.092	0.773	382
Qwen7B	0.412	0.102	0.164	0.759	382
All	0.390	0.355	0.279	0.604	382
IQ2					
LLaMa70B	0.265	0.137	0.181	0.703	374
LLaMa8B	0.318	0.695	0.436	0.290	374
ModBert	0.399	0.634	0.490	0.572	2393
OLMo32B	0.273	0.092	0.137	0.736	374
OLMo7B	0.345	0.977	0.510	0.000	374
Qwen32B	0.167	0.038	0.062	0.743	374
Qwen7B	0.369	0.290	0.325	0.693	374
All	0.305	0.409	0.306	0.534	374
WHoW					
LLaMa70B	0.312	0.159	0.211	0.721	364
LLaMa8B	0.349	0.778	0.482	0.343	364
ModBert	0.418	0.626	0.502	0.597	1816
OLMo32B	0.455	0.079	0.135	0.779	364
OLMo7B	0.349	1.000	0.517	0.025	364
Qwen32B	0.296	0.063	0.105	0.762	364
Qwen7B	0.370	0.294	0.327	0.697	364
All	0.364	0.428	0.326	0.561	364
All datasets					
LLaMa70B	0.253	0.197	0.194	0.659	2000
LLaMa8B	0.313	0.677	0.425	0.388	2000
OLMo32B	0.285	0.149	0.162	0.716	1994
ModBert	0.400	0.555	0.478	0.609	2000
OLMo7B	0.316	0.986	0.476	0.007	2000
Qwen32B	0.236	0.105	0.121	0.728	2000
Qwen7B	0.296	0.229	0.242	0.686	2000
All	0.300	0.485	0.300	0.542	2000

Table 4: Classifier performance for facilitator prediction using positive-class precision, recall and F1 ($F1_p$), as well as negative-class F1 ($F1_n$). Rows are struck through when either $F1_p$ or $F1_n$ collapses (< 0.15). Best non-collapsed $F1_p$ and $F1_n$ values for each dataset split are highlighted in bold. “All” rows report macro averages across models.

successful at identifying when they *would* (poor $F1_p$) mirroring the fact that humans are more certain of when not to intervene (§4.2).

Finding 3. *Deciding when not to facilitate is easy both for humans and LLMs. Deciding when to facilitate, is not.*

Notably, LLM behavior in facilitation is polarized: some models exhibit a pervasive tendency to constantly facilitate (reflected by extremely low $F1_n$), a pattern documented in prior work (Tsirmpas et al., 2026). Conversely, we observe that other models rarely intervene (evidenced by extremely low $F1_p$ —both crossed out in Table 4), a behavior that has been largely overlooked in the existing literature. We also find that the ModernBert classifiers can generally compete with LLMs in this task, and seem much more consistent, outperforming them in three out of the five datasets, and scoring the highest $F1_p$ scores across all datasets.⁹ We posit that overall, these classifiers may be a much more scalable, reliable and viable solution compared to LLMs for the task, even though they have been overlooked in current literature.¹⁰

Finding 4. *Out-of-the-box LLMs with generic prompting are generally not consistent in deciding when to facilitate, in contrast to traditional EO classifiers.*

5.3 Can we do better?

One way of establishing a ceiling for the performance of automated intervention models is exploring whether these models can distinguish between facilitative and non-facilitative comments in the first place. The performance of the models in this task should give us an indication of how well they understand the concept of facilitation, as well as how well-defined the task is in relevant NLP datasets. We thus explore “*facilitator detection*” a Natural Language Understanding (NLU) task which aims to identify whether a comment was made by a facilitator (Def. 2). This task has been explored exactly once by Shiota et al. (2018), although the authors used hand-crafted features instead of relying exclusively on the language used by the facilitators, and used only role-play focused scenarios instead of real-life facilitation ones.

Table 5 presents the performance of these same models in the task of facilitation detection. While we observe instances of strong performance—such as Qwen32B achieving high scores on CeRI ($F1_p=0.904$, $F1_n=0.971$)—the models generally do not score reliably across diverse datasets (e.g., LLaMa70B in WT yields $F1_p=0.500$). This suggests that the fundamental cause of under-performance is less a confusion with the traditional moderator role and more a structural issue with the datasets themselves. Prior work utilizing traditional ML (Shiota et al., 2018), LLMs, and EO classifiers shows similar challenges on other tasks (Chen et al., 2025), indicating that this finding is not merely a result

⁹There technically exists ways of boosting the performance of LLM classifiers, but this is outside the scope of this work—see App. C.1.

¹⁰Another notable advantage of these models is that we can tune the precision-recall tradeoff by adjusting the probability threshold over which the model determines that an intervention should happen, thus configuring how proactive or cautious they are—see App. C.2.3.

Model	Pr.	Rec.	F1 _p	F1 _n	Sup.
CeRI					
LLaMa70B	0.641	0.949	0.765	0.903	335
LLaMa8B	0.550	0.899	0.683	0.857	335
ModBert	0.571	0.763	0.654	0.890	152
OLMo32B	0.845	0.899	0.871	0.959	335
OLMo7B	0.345	0.367	0.356	0.793	335
Qwen32B	0.910	0.899	0.904	0.971	335
Qwen7B	0.828	0.608	0.701	0.923	335
All	0.670	0.769	0.705	0.899	335
WT					
LLaMa70B	0.410	0.641	0.500	0.500	336
LLaMa8B	0.413	0.489	0.448	0.591	336
ModBert	0.509	0.780	0.616	0.675	305
OLMo32B	0.413	0.382	0.397	0.638	336
OLMo7B	0.342	0.588	0.433	0.361	336
Qwen32B	0.509	0.214	0.301	0.733	336
Qwen7B	0.476	0.153	0.231	0.733	336
All	0.439	0.464	0.418	0.604	336
Fora					
LLaMa70B	0.571	0.830	0.677	0.753	382
LLaMa8B	0.568	0.741	0.643	0.755	382
ModBert	0.469	0.618	0.533	0.687	2736
OLMo32B	0.738	0.563	0.639	0.837	382
OLMo7B	0.215	0.404	0.440	0.695	382
Qwen32B	0.824	0.556	0.664	0.859	382
Qwen7B	0.703	0.333	0.452	0.807	382
All	0.584	0.535	0.535	0.770	382
IQ2					
LLaMa70B	0.418	0.774	0.543	0.643	374
LLaMa8B	0.394	0.617	0.481	0.662	374
ModBert	0.550	0.681	0.608	0.724	2515
OLMo32B	0.709	0.487	0.577	0.852	374
OLMo7B	0.256	0.174	0.207	0.724	374
Qwen32B	0.824	0.530	0.646	0.880	374
Qwen7B	0.826	0.165	0.275	0.836	374
All	0.568	0.490	0.477	0.760	374
WHoW					
LLaMa70B	0.476	0.901	0.623	0.651	364
LLaMa8B	0.449	0.661	0.535	0.676	364
ModBert	0.507	0.617	0.557	0.696	1869
OLMo32B	0.699	0.479	0.569	0.832	364
OLMo7B	0.274	0.140	0.186	0.727	364
Qwen32B	0.814	0.471	0.597	0.857	364
Qwen7B	0.733	0.182	0.291	0.815	364
All	0.565	0.493	0.480	0.751	364
All datasets					
LLaMa70B	0.419	0.682	0.518	0.740	2000
LLaMa8B	0.396	0.568	0.465	0.756	2000
ModBert	0.521	0.692	0.594	0.734	7577
OLMo32B	0.567	0.468	0.509	0.852	2000
OLMo7B	0.239	0.229	0.220	0.714	2000
Qwen32B	0.647	0.445	0.519	0.883	2000
Qwen7B	0.594	0.240	0.325	0.852	2000
All	0.483	0.475	0.450	0.790	2000

Table 5: Classifier performance for facilitator detection using positive-class precision, recall and F1 (F1_p), as well as negative-class F1 (F1_n). Rows are struck through when either F1_p or F1_n collapses (< 0.15). Best non-collapsed F1_p and F1_n values for each dataset split are highlighted in bold. “All” rows report macro averages across models.

of our specific labeling scheme or dataset selection.¹¹

¹¹Schroeder et al. (2024) feature F1 scores in the 0.7-0.9 range, significantly outperforming our own models. However, we were not able to replicate these results given their EO model setup. While the authors acknowledged our requests for clarifications in their training and evaluation setup, we have not received any.

Thus, significant, generalizable improvements can only be realized by incorporating larger professional facilitation datasets, where interventions can be *reliably defined and distinguished* from non-facilitative comments, or by developing larger-scale corpora that contain annotated *facilitative comments* (Def. 3)—an extremely difficult task given the inherent subjectivity of the task (§4).

Finding 5. *By defining facilitation as speech produced by facilitators, we handicap the potential of predictive facilitative models.*

6 Conclusion

We investigated how the timing of humans and LLMs differs in facilitative interventions. By conducting a literature review, we isolated three main definitions of facilitative interventions. We then conducted a survey task using facilitative social science experts on a standardized, aggregated corpus of all relevant facilitation datasets, finding that human facilitators tend to adopt a cautious approach, are more confident when deciding not to intervene than when deciding to intervene, and may disagree because they focus on different spans of the discussion. We are the first to conduct extensive experiments on facilitative intervention prediction using LLMs, finding that they tend to over-intervene, skew towards positive reinforcement when the situation is ambiguous, and remain inconsistent across facilitation decisions, while simpler ModernBERT classifiers trained on existing datasets can outperform them. We also discovered that the way facilitative interventions are usually defined in the literature may place a ceiling on the performance of classifiers.

Despite optimism that LLMs would revolutionize facilitation, our findings suggest that progress may be incremental, not transformative—LLMs can not meaningfully facilitate until we move beyond limited datasets. Previous approaches—such as focusing on escalation or general moderation—provided large, scalable data, but only captured surface-level behaviors. They failed to give us a true understanding of how human facilitators operate, a complex and open question that can not be answered without annotated data that captures genuine facilitative dynamics.

Limitations

This work is not without limitations. With additional resources, we could have incorporated more survey or, given that we understood the task’s subjectivity more adequately, even annotation schemata (e.g., sentence-by-sentence, apart from the excerpt) and explored a wider range of prompting settings. Also the opposite is true; a possible criticism of our decision is that we do not provide the full discussion thread, which in some cases could be extremely large and impractical to include in its entirety. Therefore, this organic limitation led to us to choose a middle ground, that is providing excerpts of the discussions. Facilitation decisions in real-world settings

are often shaped by long-term interaction dynamics, participant history, community norms, and evolving conversational context, which may not be fully captured in the excerpts. Furthermore, because providing rationales was optional for participants, we collected very few, resulting in the loss of a potentially valuable resource for interpretation—this information is valuable and should be incorporated in future tasks concerning facilitation. Moreover, the intervention strategies examined here are only two: positive and negative reinforcement. Ideally, the task should be conducted without the use of different intervention labels or very fine-grained labels, as prior work has shown that different domains call for different facilitative tactic taxonomies (Park et al., 2012; Chen et al., 2025; Schroeder et al., 2024). The decision of including the distinction between negative and positive reinforcement was motivated by the fact that the participants were over-focusing on the moderation part during the refinement stage of the task.

Additionally, although early work attempted to estimate interventions in terms of external metrics such as toxicity (Hua et al., 2018; Chang and Danescu-Niculescu-Mizil, 2019; Yuan and Singh, 2023), recent work has shown that facilitators may have different objectives depending on the form of discussion, which may require the use of different evaluation metrics for each domain (Korre et al., 2025). Since our work attempts to generalize interventions across multiple domains, we can not rely on external signals, but only on whether an intervention would (or did) occur.

Finally, the datasets included in PEFK are predominantly English-language and drawn from specific online or institutional discussion settings. As a result, the findings may not generalize to multilingual environments and culturally distinct moderation norms.

Ethical Considerations

All experiments and data have passed through official approval and were overseen by an Ethics Committee. The participants were selected via formal recruitment methods based on familiarity with the task and field, and were compensated fairly with a monthly salary of 1000€. ¹² They signed informed consent forms before beginning the survey, were informed and consented to being exposed to potentially offensive or inappropriate content, and were allowed to skip uncomfortable tasks at any point, with no required justification. All necessary data were anonymized both during data collection and in the final, presented artifacts, while non-essential information was destroyed with the conclusion of the survey in accordance with GDPR (European Union, 2016) and the EU AI Act (European Union, 2024). Participant demographic data was not collected, since it was deemed unnecessary for the purposes of this study.

In order to help research in the field of facilitation,

¹²Specific information on posted requirements and the recruitment process will be made available after review, since they would compromise anonymity.

we openly release the code that constructs PEFK under a GPLv3 license. All datasets used in PEFK are obtained from the sources provided by the respective authors and are consistent with intended use (academic research). We do not distribute the corpus separately, and thus no additional data license is applicable. The licenses of the original distribution apply. The Fora dataset (Schroeder et al., 2024) is not publicly available because it may contain personally identifying information; researchers should contact the authors of Fora for access. Finally, we use exclusively open-source models, libraries and LLMs, in order to ensure reproducibility, protect user data, and comply with relevant regulations (see App. C.1).

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We used LLMs for code generation and internal documentation in some instances. All outputs have been verified by the authors.

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A Acronyms

NLP	Natural Language Processing
NLU	Natural Language Understanding
ML	Machine Learning
RQ	Research Question
PEFK	Prosocial and Effective Facilitation in Konversations
EO	Encoder-Only
IAA	Inter-Annotator Agreement
UMOD	User Moderation
IQ2	Intelligence Squared
CeRI	Cornell eRulemaking Initiative
WHoW	Why-How-Who
WT	WikiTactics
CoT	Chain of Thought

Name	Type	Description
conv_id	string	The discussion’s ID. Comments under the same discussion refer to the same discussion ID.
message_id	string	The message’s (comment’s) unique ID.
reply_to	string	The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it’s the Original Post (OP)).
user	string	Username or hash of the user that posted the comment
is_moderator	bool	Whether the user is a facilitator.
moderation_supported	bool	True if the moderation labels are directly computed from the original dataset
escalated	bool	A discussion-level measure denoting discussions which have been de-railed
escalation_supported	bool	True if the escalation labels are directly computed from the original dataset
text	string	The contents of the comment
dataset	string	The dataset from which this comment originated from
notes	JSON	A dictionary holding notable dataset-specific information
toxicity	float	The “toxicity” score given to the comment by the Perspective API
severe_toxicity	float	The “severe toxicity” score given to the comment by the Perspective API

Table 6: The full set of columns used for each comment in the [PEFK](#) dataset.

B Dataset details

B.1 Preprocessing

General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wikitactics and wikiconv as well as wikidisputes and wikiconv.
 - There may be duplicate discussions between wikidisputes and wikitactics, but we allow them since they feature complementary information.

Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers ([Hua et al., 2018](#)), and select only discussions which have at least two comments by different users.
 - Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their user_id is always set to 0 and their username is of the form 211.111.111.xxx. We consider each such username to be a separate user.
 - Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.

- We only include English comments in the final dataset. We use a small, efficient library (py3langid) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

Wikitactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
 - Asking questions
 - Coordinating edits
 - Providing clarification
 - Suggesting a compromise
 - Contextualisation
- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper. They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as “Conceding/recanting” and “I don’t know”.

Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% participant agreement (measured as entropy in the original dataset).

CMV-AWRY2

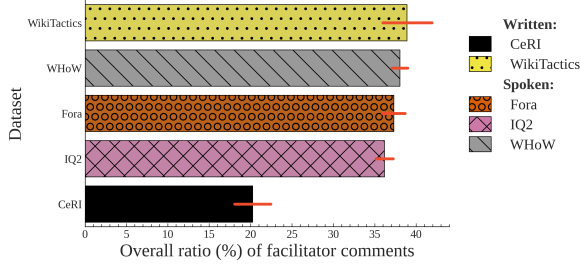


Figure 5: Percentage of comments made by professional facilitators (see Def. 2) for each of the sub-datasets of PEFK. Bars denote 95% confidence intervals. Despite the different kinds of facilitators in online platforms and various real-life-discussion domains, the overall rate of facilitation remains similar across datasets.

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted (“[deleted]”) comments.

B.2 Schema

Table 6 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the `is_moderator` column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

B.3 Exploratory Data Analysis

We explore some initial frequencies with regard to facilitator comments in PEFK, comparing its oral vs. written subsets, as well as vocabulary discrepancies.

Oral vs. written datasets in PEFK There currently exists a rift in facilitation research; while some previous work concerns itself with forums and online discussions (Park et al., 2012), others are exclusively focused on live, real-life, oral discussions (Schroeder et al., 2024; Chen et al., 2025). This rift is an issue for facilitation research, since oral and online written discussions feature fundamentally different kinds of facilitation interventions, as shown by stark differences in proposed facilitation schemas between online forums (CeRI, UMOD), and real-life discussions (WHoW, IQ2, Fora). Additionally, the data themselves are different; posts and comments are self-contained, while transcripts regularly feature interruptions, participants talking in parallel (see App. B.3), and linguistic features that are necessary for coordination (e.g., “Thank you.”, “I agree.”). It is worth noting that in our case, all written discussions are asynchronous (the participants may join and leave at any time), while all oral discussions are synchronous (the participants are all active at the same time).

Figure 6 shows some of the structural differences of oral and written datasets. oral datasets feature a significantly

larger number of comments per discussion, since instances where the participants talk over each other, or interrupt the other are transcribed as separate comments. Additionally, oral datasets feature a large amount of comments with very few words, since a lot of utterances are short but necessary for social co-ordination (e.g., “Thank you”, “Yes”, “I agree”). Figure 5 shows the percentage of comments made by facilitators for each of our datasets. In almost all datasets, a third of the comments are made by facilitators.¹³

Vocabulary discrepancies in PEFK datasets. We also the data to include only messages written by moderators, then cleaned the text by removing URLs, punctuation, and other noise. After preprocessing, we used a TF-IDF vectorization approach with n-grams (bigrams and trigrams) to extract phrases. Crucially, the TF-IDF model was trained on all datasets combined, so that common, generic phrases were downweighted while phrases that are more distinctive to each dataset were emphasized. We then computed the average TF-IDF score of each phrase within each dataset and selected the top-ranking ones as representative expressions of moderator language in that context. The results can be found in Figure 7.

The most important phrases in CeRI suggest a formal, policy-oriented environment with an emphasis on structured participation and regulatory discussion (e.g., references to “proposed rule” and “welcome regulations”). In *fora*, the language is more conversational and facilitative, with phrases such as “great thank”, “feel free”, and “awesome thank” indicating a more informal moderation style. The IQ2 dataset reflects the debate setting, with phrases like “ladies gentlemen” and “arguing motion”. In UMOD, moderators engage more in argumentation and clarification, as seen in phrases like “change view”, “missing point” reminiscing more facilitative settings and suggesting a focus on reasoning and discussion quality. The WHoW dataset appears similar to moderated discussions or panels, with phrases like “question sir” and “answer question” indicating turn-taking and audience interaction. Finally, WikiTactics shows a highly procedural and rule-based style, with phrases such as “reliable sources”, “original research” and “talk page” reflecting the norms and policies of collaborative editing environments like Wikipedia.

C Models

C.1 LLMs

We use only open-source models, as proprietary models such as OpenAI’s GPT-5 yield much less reproducible results (Bisbee et al., 2024). Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings (Tsirmpas et al., 2026). It is also worth noting that

¹³We do not include the UMOD dataset, since there is no established ground-truth (see App. B.1).

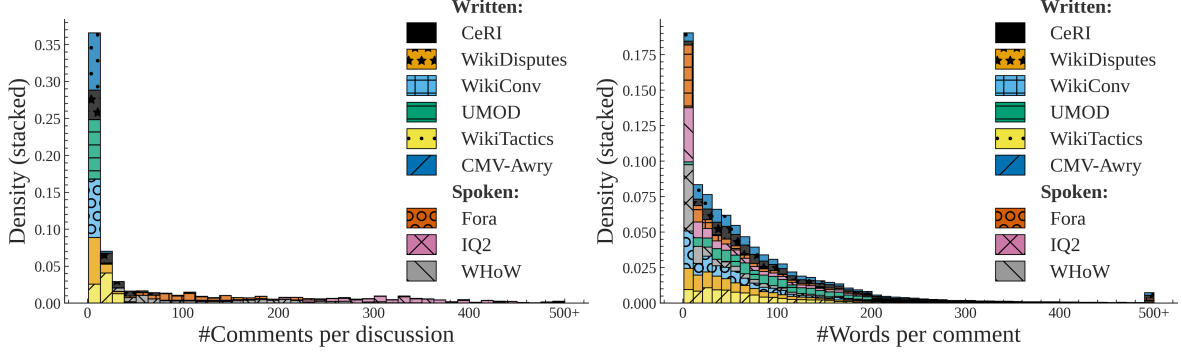


Figure 6: **Left:** Comments per discussion on the PEFK dataset. Note the significant increase in comments made in transcribed datasets such as Fora and IQ2. This deviation is explained by the fact that a single human utterance is regularly broken up to many comments in order to simulate interruptions or participants talking over each other. **Right:** Number of words per comment in all discussions in PEFK, grouped by original dataset. Comments across written datasets follow the same distribution, while oral datasets feature a significant number of comments with very few words.

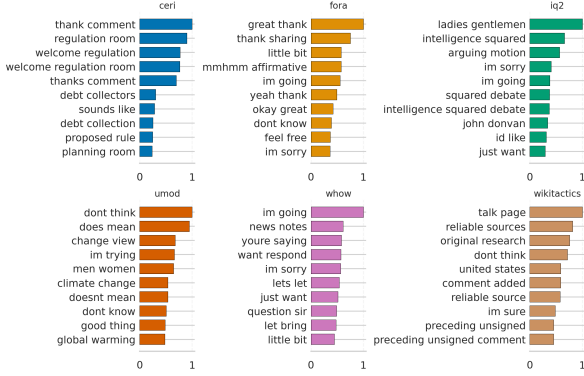


Figure 7: Most common moderator bigrams in PEFK per dataset. Function words are excluded.

since a part of our dataset (Fora) is not licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation (European Union, 2024, 2016).

Proprietary LLMs such as the OpenAI GPT family may perform better, but cost several orders of magnitude more, and are thus not a realistic solution (Tsirmpas et al., 2026). While the performance of LLM classifiers could be reasonably increased by providing few-shot examples or finetuning, these approaches would significantly bias the model since no concrete definition of facilitation exists (Korre et al., 2025), and the examples would have to be different for each different discussion domain in PEFK—which goes against our motivation of discovering patterns between oral and written datasets.

Table 7 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the unsloth huggingface repository.

Family	Version	#Params
OLMo	2 (0325)	32B
Qwen	2.5	32B
LLaMa	3.3	70B
OLMo	3	7B
Qwen	2.5	7B
LLaMa	3.1	8B

Table 7: LLMs used in this study.

C.2 Transformers

C.2.1 Training details

We construct our training and evaluation data from a unified corpus of threaded discussions, where each instance corresponds to a target comment augmented with its conversational context. For every target comment, we retrieve up to $K = 3$ preceding comments along the reply chain. Each comment is truncated to a maximum of 3000 characters to control sequence length. The input is formatted using XML-style tags, where context comments are wrapped with `<CTX>` and the target comment with `<TGT>`. The final sequence is formed by concatenating context comments in chronological order followed by the target comment. We consider binary classification tasks (e.g., `is_moderator`, `should_intervene`), with labels cast to floating point values.

We fine-tune a pretrained transformer encoder (ModernModBert-large) with an added classification head that outputs a single logit. The task is formulated as a single-label instance of multi-label classification, enabling the use of a sigmoid activation.

Training is conducted using the Hugging Face Trainer framework with several modifications. We optimize the model using binary cross-entropy with logits:

$$\mathcal{L} = \text{BCEWithLogitsLoss}(\mathbf{z}, \mathbf{y}; \text{pos_weight}),$$

where $\mathbf{z}$ are the predicted logits and $\mathbf{y}$ are the ground-truth labels. To account for class imbalance, the positive class is weighted as. We freeze the base transformer parameters and train only the classification head in order to reduce computational cost and limit overfitting.

To improve computational efficiency, we employ a bucketed batching strategy that groups instances of similar length (approximated in characters) into batches of size 64. This reduces padding and improves throughput. Batches are shuffled at the bucket level at each epoch.

Training is performed for up to 80 epochs, with evaluation, logging, and checkpointing conducted once per epoch. The best model is selected based on validation loss. We apply early stopping with a warm-up phase: monitoring begins after one evaluation step, and training stops if validation loss does not improve by at least $\delta = 0.001$ for 6 consecutive evaluation steps. Model performance is evaluated using accuracy and binary F1 scores.

All experiments are implemented in PyTorch using the Hugging Face Transformers library. Data loading uses 4 worker processes, and GPU acceleration is utilized when available. We fix the random seed to 42 to ensure reproducibility.

C.2.2 One model to rule them all?

Since comments in the written and oral datasets are fundamentally different (§B.3), we would be tempted to train different models for each. This decision is not obvious however, since it is likely that the model can internally distinguish between oral and written speech. If that occurs, the differences between the texts should not negatively affect the model; in fact, training the model on all datasets could allow it to transfer knowledge between the two text modalities. Table 8 shows the performance of the same model when trained only on oral and written datasets, compared to when trained on the all datasets at once. We find that the models perform better when trained on different splits.

Split	Dataset	Precision	Recall	F1
Written	CeRI	0.320	0.490	0.387
	Wikitactics	0.407	0.717	0.520
oral	Fora	0.409	0.631	0.497
	IQ2	0.399	0.634	0.490
	WHoW	0.418	0.626	0.502
All	CeRI	0.292	0.172	0.216
	Fora	0.410	0.664	0.507
	IQ2	0.406	0.623	0.491
	WHoW	0.422	0.642	0.509
	Wikitactics	0.456	0.364	0.405

Table 8: Summary of classifier performance depending on the training dataset splits on the facilitator prediction task. We compare models trained on all available data with models trained separately on written and oral datasets. Performance is generally similar.

C.2.3 Decision thresholds

An advantage of using EO classifiers instead of LLMs is that we can quantitatively alter the performance of the classifier by tuning the thresholds required for a comment to trigger an intervention. By making the model more conservative (increasing the threshold), we could hope that we can avoid unnecessary interventions, while making it less conservative (decreasing the threshold) we could make it more proactive.

Fig. 8 shows that the classifier has an inherently low precision score which is largely invariant to the selected threshold. On the other hand, recall falls off rapidly, which incentivises us to use a more liberal approach.

D Human survey procedure

D.1 Facilitative tactics

Before the main survey on facilitative intervention, all participants were extensively trained on currently established facilitative tactics. This task aimed to familiarize the participants with what constitutes a facilitative intervention, as well as how professional facilitators operate in real-life scenarios.

Specifically, we split the participants into three groups, each of which was assigned a different taxonomy of facilitative interventions (Chen et al., 2025; Schroeder et al., 2024; Park et al., 2012). The participants were tasked with classifying 830 facilitated comments from a sample of continuous discussions in PEFK. The guidelines and results of this annotation are not shown in this study, but they are available upon request.

D.2 Positive vs. negative reinforcement

The distinction between positive/negative reinforcement is present to suggested facilitative tactics taxonomies (Table 2), including prior work such as Fora (Schroeder et al., 2024) WHoW (Chen et al., 2025), and CeRI (Park et al., 2012). We conservatively estimate whether the tactics described in each taxonomy correspond to our definitions of positive and negative reinforcement.

The facilitative taxonomy proposed in Fora does not include any tactics that could be unambiguously considered as negative reinforcement, which is expected, since facilitators deployed in the relevant study were supposed to only act as coordinators. Notably, several tactics may simultaneously exhibit both positive and negative characteristics across all taxonomies, depending on context and the exact implementation used by the facilitator. We therefore use these taxonomies only as guidance during annotation, and refer the reader to the original works for a more comprehensive description and analysis of facilitation strategies.

D.3 Label Transformation

To efficiently calculate intra-participant consistency, we transform the fine-grained, ordinal label per type adopting a multiclass schema. We define a threshold of ≥ 3 and then using Table 10 we relabel the items.

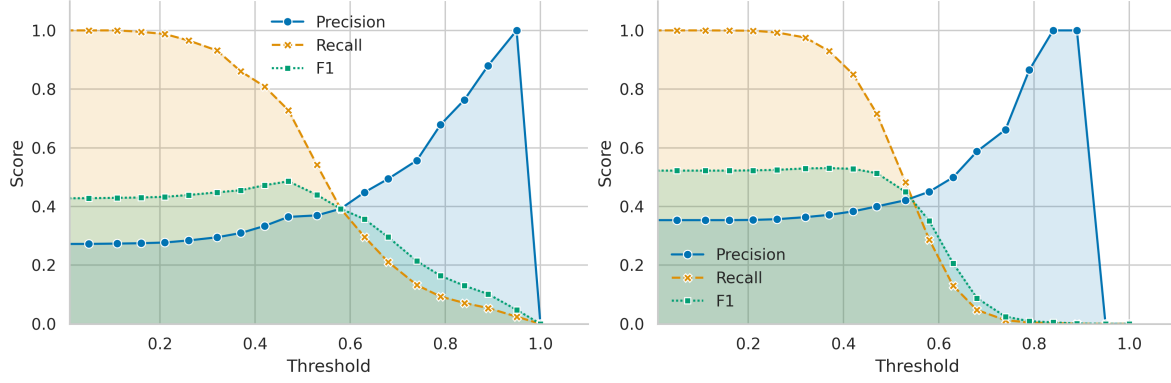


Figure 8: Classifier performance (**left**: written datasets, **right**: oral datasets) when tuning the decision threshold on the test-set. The higher the threshold, the more conservative the classifier is encouraged to become when predicting a facilitator intervention.

D.4 Are facilitative tendencies dataset-dependent?

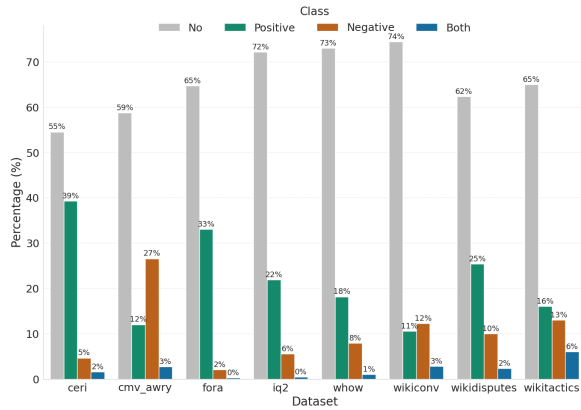


Figure 9: Label distributions of human participants per dataset.

Facilitative tendencies generally follow a conservative pattern, with no reinforcement selected most often. Despite the differences between the datasets (forums, debates, and radio podcasts), we observe similar trends in the choice between positive and negative reinforcement, with positive reinforcement being more common overall. Two exceptions are *cmv_awry* and *wikiconv*, where negative interventions are selected more often than positive ones, by 15% and 1%, respectively (Fig. 9).

D.5 Participant consistency

To evaluate the consistency of individual participants, we measured intra-participant agreement using purposefully repeated instances of identical discussions.

Agreement was computed as a strict match in the labeling of the same instance (duplicated pair). For each duplicated pair, agreement was assigned a value of 1 if both labels were identical and 0 otherwise in order to calculate a simple pairwise agreement. Hence, the final intra-participant agreement for each participant was then calculated as the mean of these binary agreement scores. Since our classes are heavily skewed and imbalanced, traditional agreement metrics tend to collapse,

showing extremely low scores even for almost-100% agreement (Röttger et al., 2022; Feinstein and Cicchetti, 1990). For the purposes of simplicity, we thus run our analysis in this part with simple percentage agreement.

We report results in Table 11. Most participants demonstrated near-perfect consistency, with several achieving 100% agreement. Participant A3 is the only participant who showed reduced consistency (66.67%), suggesting potential variability in interpretation or labeling behavior.

E Survey guidelines

E.1 Human guidelines

Survey Guidelines: Type of Reinforcement Needed in Comment Threads

Purpose of the Task

These guidelines explain how to rate comments in conversation threads using three parallel rating columns:

- Positive Reinforcement (1–5)
- Negative Reinforcement (1–5)
- No Reinforcement Needed (1–5)

Your job is to read each comment in its conversational context and assign a strength/certainty rating (1–5) for each category, reflecting how that category applies. All fields need to be filled and the categories are not necessarily mutually exclusive.

Even though your intervention will be delivered at the end of the thread, the reinforcement can refer to any point in the conversation, not just at the end. It can refer to something said earlier, not just the last comment.

You are not being asked whether that conversation correlates to positive or negative reinforcement, but rather whether you, personally, would have used positive or negative reinforcement at a given point. Keep in mind that too many interventions on the moderator’s part may result in an adverse reaction from the participants; no one wants to be constantly told what to say or how to say it.

Important Notes

- You must add the rationale behind your response in the rationale/comments field when:

Taxonomy & Tactic	Type
Fora	
Express appreciation	+
Express agreement or affirmation	+
Open invitation to participants	
Specific invitation to participant	
Model examples	
Follow-up question	
Make connections	
WHoW	
Informational	
Coordinative	-
Social	+
Probing	
Confronting	
Instruction	-
Interpretation	
Supplement	
CeRI	
Welcoming	+
Encouragement / appreciation	+
Maintaining civil discourse	-
Explaining why comment not removed	
Indicating irrelevant/off-point comments	-
Pointing to relevant information	
Pointing out effective commenting	+
Asking for more information/sources	
Asking for solutions/alternatives	
Encourage engagement with others	
Pose question to community	

Table 9: Facilitation taxonomies (bold) with associated tactics (normal text) and reinforcement type. +: positive reinforcement, -: negative reinforcement, empty if the tactic does not clearly correspond to any category.

Positive	Negative	Neither	Final
X	X	X	None (0)
✓	X	X	Positive (1)
X	✓	X	Negative (2)
X	X	✓	None (0)
✓	✓	X	Both (3)
✓	X	✓	Positive (1)
X	✓	✓	Negative (2)
✓	✓	✓	Both (3)

Table 10: Reinforcement label transformations.

- your positive score > 3, or
- your negative score > 3, or
- you encounter a particular case (e.g., an instance which could fit both negative and positive reinforcement categories).
- If you encounter data that is malformed or noisy to the point that it renders comprehension difficult, flag the instance using the malformed field.

What Each Label Means

Positive Reinforcement

Positive reinforcement is defined as an intervention whose purpose is to encourage a specific behavior. For

participant	Agreement (%)
A2	100.00
A3	66.67
A4	88.89
A5	100.00
A6	100.00
A7	100.00
A8	100.00
A9	100.00
A10	100.00
A11	100.00

Table 11: Intra-participant percentage agreement (consistency) over all subdatasets of PEFK.

example, a moderator may ask people to participate or ask a person to elaborate. Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors (e.g., “I will not tolerate any further discussion on this topic. It’s been decided and we’re moving on.”).

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action. Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., “I’ve received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.”).

Rate this when the comment reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where some participants dominate the conversation, or when the conversation starts to go off topic. Other examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the

conversation

No Reinforcement Needed

Rate this when the comment is neutral, unremarkable, or does not affect the tone or direction of the conversation. This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but non-notable contributions

How to Make Decisions

Look at Intent and Impact

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context

Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions

The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent

Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

E.2 LLM prompts

E.2.1 Yes/no intervention

You are an expert specializing in online moderation and facilitation. You will be presented with a comment and the comments made before it. Choose whether you decide to intervene at this point in the discussion.

Key Rules:

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether you would intervene in this discussion as a facilitator on a scale of 1-5, where 1 is "I'm sure I don't need to intervene", and 5 is "I'm sure I need to intervene"

E.2.2 Positive vs. negative reinforcement

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

The Core Rating Task

For every comment, you must write out a response of 1 (uncertain) to 5 (certain):

Positive Reinforcement

Defined as an intervention whose purpose is to encourage a specific behavior. For example, a moderator may ask people to participate, ask a person to elaborate etc.

Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors (e.g., "I will not tolerate any further discussion on this topic. It's been decided and we're moving on.").

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action.

Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., "I've received multiple complaints about your behavior on this forum. Please refrain from making personal attacks and inflammatory comments.").

Rate this when the comment reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where some participants dominate the conversation, or when the conversation starts to go off topic.

Other examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

No Reinforcement

Rate this when the comment is neutral, unremarkable, or does not affect the tone or direction of the conversation.

This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but non-notable contributions

Key Rules:**Look at Intent and Impact**

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context

Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions

The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent

Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output

Respond whether you would intervene in this discussion as a facilitator.

Always respond with the following format and nothing else:

Positive: 1-5

Negative: 1-5

No Reinforcement: 1-5

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether the comment you are reading (TGT) is a comment made by a moderator/facilitator. Write 0 if it was not made by a facilitator, and 1 if it was made by a facilitator. Do not write anything else, only the number.

E.2.3 Single-comment facilitation prediction

You are an expert moderator/facilitator. You will be presented with a comment (TGT) and the comments made before it (CTX, one for each comment in turn). Choose whether you decide to intervene at this point in the discussion.

Key Rules:

Look at Intent and Impact Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

Use the Context Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context—and vice versa.

Judge Behavior, Not Opinions The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

Remain Consistent Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

Output Respond whether you would intervene in this discussion as a facilitator. Write 0 if you would not intervene, and 1 if you would. Do not write anything else, only the number.

E.2.4 Single-comment facilitation detection

You are an expert moderator/facilitator. You will be presented with a comment (TGT) and the comments made before it (CTX, one for each comment in turn). Choose whether the comment you are reading (TGT) is a comment made by a moderator/facilitator.

Key Rules for: